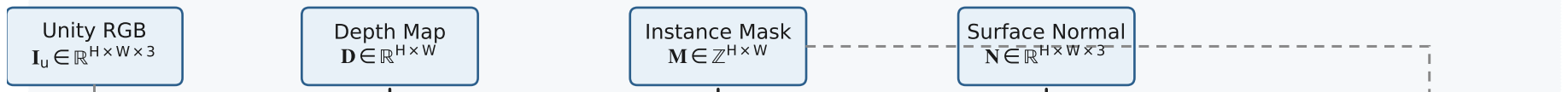


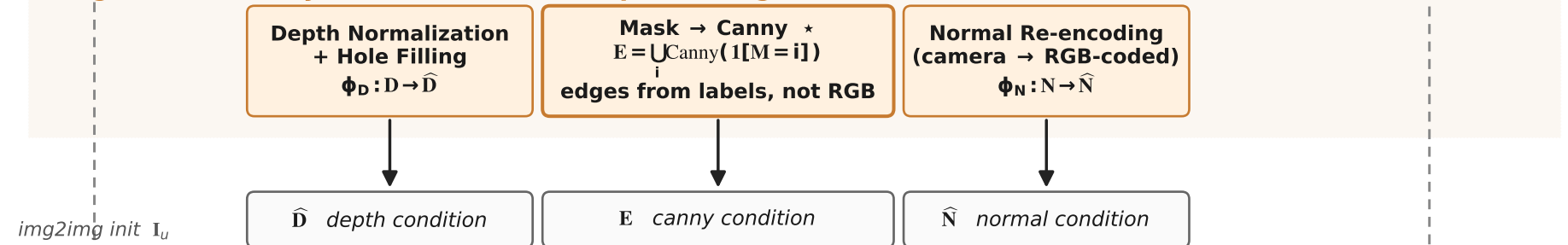
Multi-Conditional ControlNet Sim2Real Translation Pipeline

for Open-Pit Mine Blast-Pile Instance Segmentation Data Synthesis

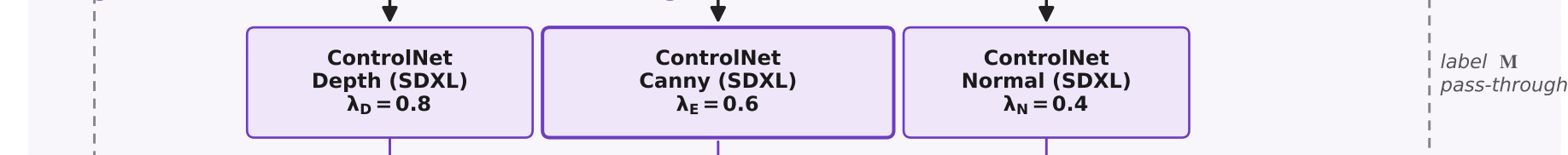
Stage A Unity Perception Synthetic Inputs



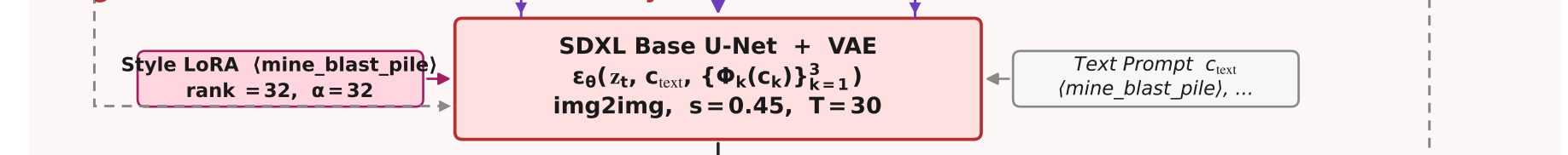
Stage B Geometry-Aware Condition Preprocessing



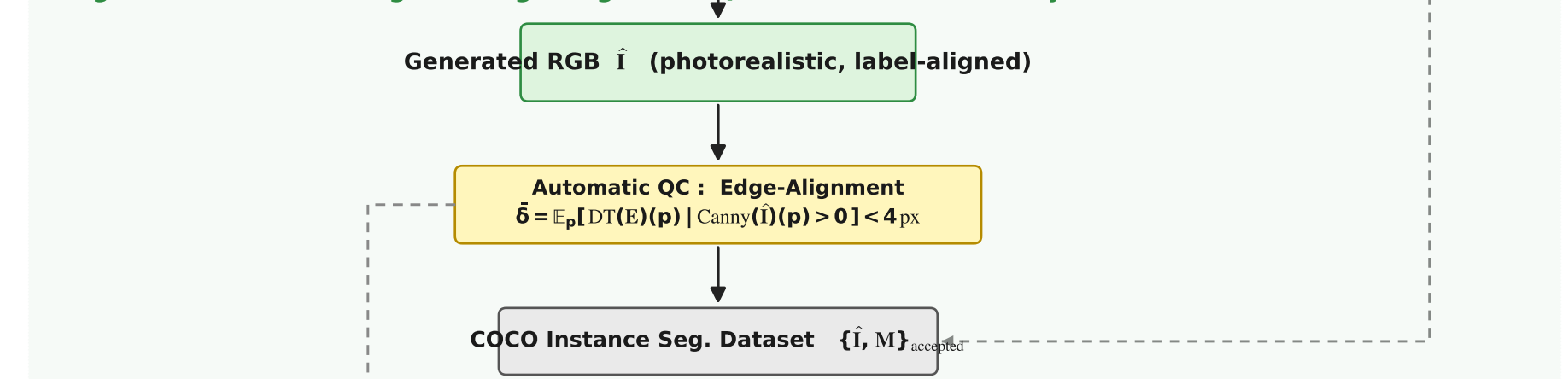
Stage D Multi-Path ControlNet Conditioning



Stage E SDXL Diffusion Backbone with Style LoRA



Stage F Generated Image + Edge-Alignment QC + Dataset Assembly



→ data flow - - -> auxiliary (init / pass-through / reject) → ControlNet zero-conv injection

★ **Key contribution** : The Canny condition E is derived directly from instance masks M , not from the RGB image — this guarantees pixel-perfect alignment between generated edges and ground-truth segmentation boundaries.